

LAPORAN PRAKTIKUM
MATA KULIAH SISTEM TERDISTRIBUSI

Disusun sebagai bentuk pemenuhan tugas pada matakuliah system terdistribusi

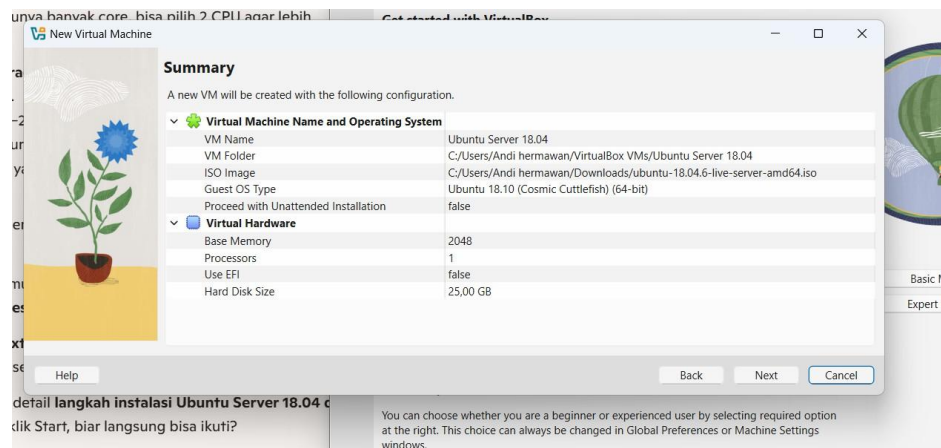


Ditulis Oleh:

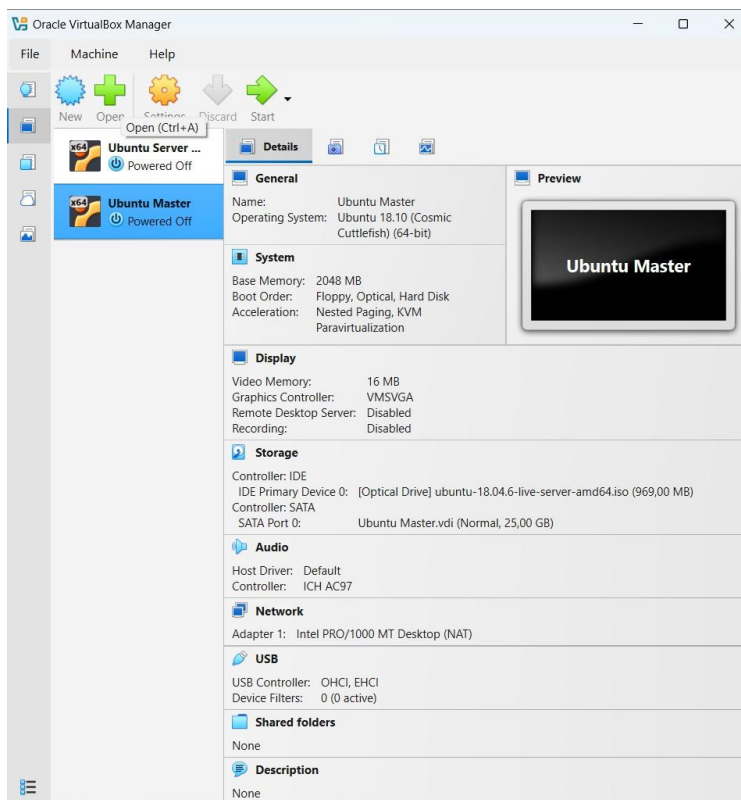
Calvin William Wijaya (10602008)

PROGRAM STUDI TEKNOLOGI REKAYASA PERANGKAT LUNAK
JURUSAN TEKNOLOGI INFORMASI DAN KOMPUTER
POLITEKNIK NEGERI SUBANG
TAHUN 2026

A. Screenshoot Pengerjaan



Berhasil instal VM pada virtual box :



Profile setup

Auto capture keyboard

Don't show

Enter the username and password you will use to log in to the system. You can configure SSH access on the next screen but a password is still needed for sudo.

Your name: Andi Hermawan

Your server's name: ubuntu-master
The name it uses when it talks to other computers.

Pick a username: andi

Choose a password: ****

Confirm your password: ****

[done]

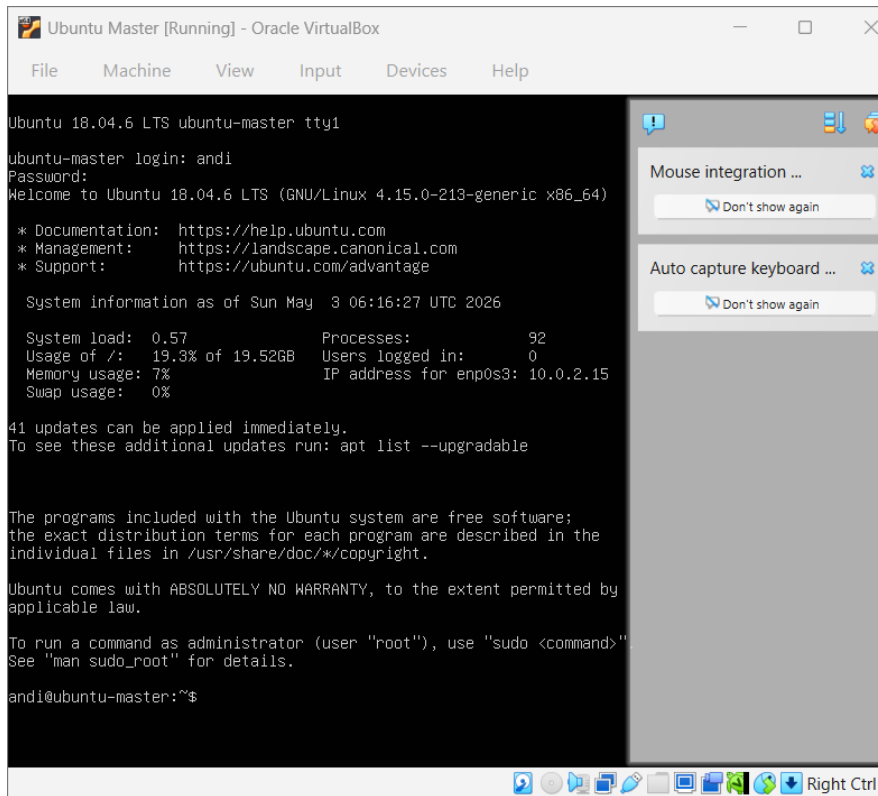
Instal Ubuntu:

Installing system

running 'curtin block-meta simple'
curtin command block-meta
removing previous storage devices
configuring disk: disk-sda
configuring partition: partition-0
configuring partition: partition-1
configuring format: format-0
configuring partition: partition-2
configuring lvm_volgroup: lvm_volgroup-0
configuring lvm_partition: lvm_partition-0
configuring format: format-1
configuring mount: mount-1
configuring mount: mount-0
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///media
configuring installed system
running '/snap/subiquity/2651/bin/subiquity-config
/snap/subiquity/2651/usr/bin/python3 true'
curtin command apt-config
curtin command in-target
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
installing kernel |

[View full log]

Instal ubuntu selesai:



```
Ubuntu Master [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Ubuntu 18.04.6 LTS ubuntu-master tty1
ubuntu-master login: andi
Password:
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 4.15.0-213-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Sun May  3 06:16:27 UTC 2026

System load:  0.57           Processes:           92
Usage of /:   19.3% of 19.52GB Users logged in:      0
Memory usage: 7%            IP address for enp0s3: 10.0.2.15
Swap usage:   0%

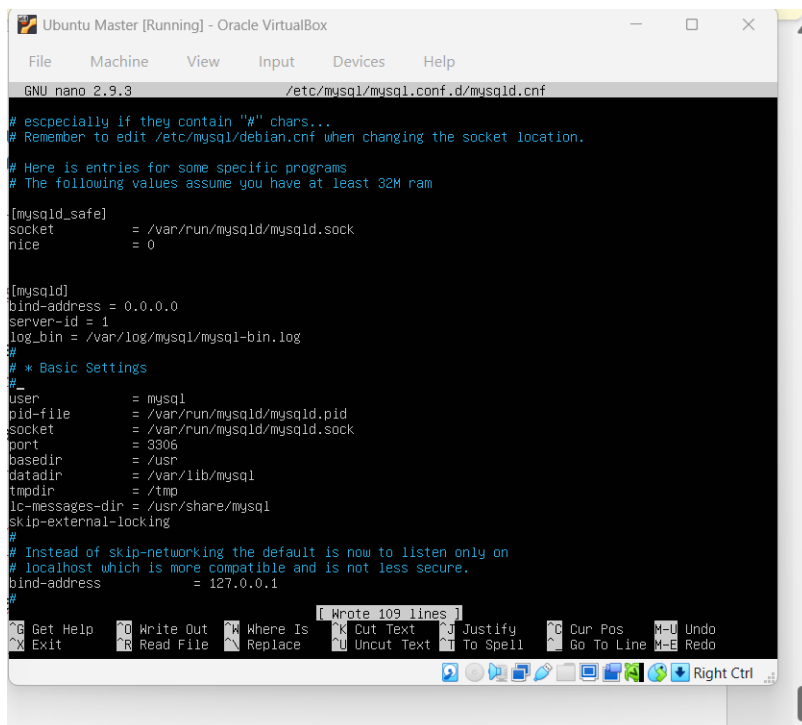
41 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>"
See "man sudo_root" for details.

andi@ubuntu-master:~$
```



```
Ubuntu Master [Running] - Oracle VirtualBox
File Machine View Input Devices Help

GNU nano 2.9.3 /etc/mysql/mysql.conf.d/mysqld.cnf
# especially if they contain "#" chars...
# Remember to edit /etc/mysql/debian.cnf when changing the socket location.
# Here is entries for some specific programs
# The following values assume you have at least 32M ram

[mysqld_safe]
socket      = /var/run/mysql/mysql.sock
nice        = 0

[mysqld]
bind-address = 0.0.0.0
server-id = 1
log_bin = /var/log/mysql/mysql-bin.log
#
# * Basic Settings
#
user        = mysql
pid-file    = /var/run/mysql/mysql.pid
socket      = /var/run/mysql/mysql.sock
port        = 3306
basedir     = /usr
datadir     = /var/lib/mysql
tmpdir      = /tmp
lc-messages-dir = /usr/share/mysql
skip-external-locking
#
# Instead of skip-networking the default is now to listen only on
# localhost which is more compatible and is not less secure.
bind-address = 127.0.0.1
#

Wrote 109 lines
Get Help Write Out Where Is Cut Text Justify Cur Pos M-U Undo
Exit Read File Replace Uncut Text To Spell Go To Line M-E Redo
```

```

andi@ubuntu-master:~$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 3
Server version: 5.7.42-0ubuntu0.18.04.1 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW MASTER STATUS;_

```

STATUS VARIABLES:

```

owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW VARIABLES LIKE 'log_bin';
+-----+-----+
| Variable_name | Value |
+-----+-----+
| log_bin       | ON    |
+-----+-----+
1 row in set (0.00 sec)

mysql> SHOW VARIABLES LIKE 'server_id';
+-----+-----+
| Variable_name | Value |
+-----+-----+
| server_id     | 1     |
+-----+-----+
1 row in set (0.00 sec)

mysql> _

```

CREATE DB:

```

mysql> CREATE DATABASE testdb;
Query OK, 1 row affected (0.01 sec)

mysql> SHOW MASTER STATUS;
+-----+-----+-----+-----+-----+
| File           | Position | Binlog_Do_DB | Binlog_Ignore_DB | Executed_Gtid_Set |
+-----+-----+-----+-----+-----+
| mysql-bin.000001 |      319 |              |                  |                  |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> _

```

Ganti server-id:

```
Ubuntu Master [Running] - Oracle VirtualBox
File Machine View Input Devices Help
GNU nano 2.9.3 /etc/mysql/mysql.conf.d/mysqld.cnf

# It has been reported that passwords should be enclosed with ticks/quot
# especially if they contain '#' chars...
# Remember to edit /etc/mysql/debian.cnf when changing the socket locati

# Here is entries for some specific programs
# The following values assume you have at least 32M ram

[mysqld_safe]
socket      = /var/run/mysqld/mysqld.sock
nice        = 0

[mysqld]
bind-address = 0.0.0.0
server-id = 2
log_bin = /var/log/mysql/mysql-bin.log
relay-log = /var/log/mysql/mysql-relay-bin.log
#
# * Basic Settings
#
user                = mysql
pid-file            = /var/run/mysqld/mysqld.pid
socket              = /var/run/mysqld/mysqld.sock
port                = 3306
basedir             = /usr
datadir             = /var/lib/mysql
tmpdir              = /tmp
lc-messages-dir     = /usr/share/mysql
skip-external-locking
#
# Instead of skip-networking the default is now to listen only on
# localhost which is more compatible and is not less secure.

andi@ubuntu-master:~$ sudo systemctl_
```

```
andi@ubuntu-master:~$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 2
Server version: 5.7.42-0ubuntu0.18.04.1-log (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW VARIABLES LIKE 'server_id';
+-----+-----+
| Variable_name | Value |
+-----+-----+
| server_id     | 2     |
+-----+-----+
1 row in set (0.00 sec)

mysql> SHOW VARIABLES LIKE 'log_bin';
+-----+-----+
| Variable_name | Value |
+-----+-----+
| log_bin       | ON    |
+-----+-----+
1 row in set (0.00 sec)

mysql>
```

Buat USER repl di MASTER:

```
mysql> CREATE USER 'repl'@'%' IDENTIFIED BY 'password123'; GRANT REPLICATION SLAVE ON *.* TO 'repl'@
'%'; FLUSH PRIVILEGES;
ERROR 1396 (HY000): Operation CREATE USER failed for 'repl'@'%'
Query OK, 0 rows affected (0.01 sec)

Query OK, 0 rows affected (0.00 sec)

mysql>
```

```
mysql> CHANGE MASTER TO
-> MASTER_HOST='10.0.2.15',
-> MASTER_USER='repl',
-> MASTER_PASSWORD='password123',
-> MASTER_LOG_FILE='mysql-bin.000001',
-> MASTER_LOG_POS=319;
Query OK, 0 rows affected, 2 warnings (0.06 sec)

mysql> START SLAVE;
```

Berhasil login SLAVE:

```
mysql> exit;
Bye
andi@ubuntu-master:~$ mysql -h 10.0.2.15 -u repl -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 23
Server version: 5.7.42-0ubuntu0.18.04.1-log (Ubuntu)

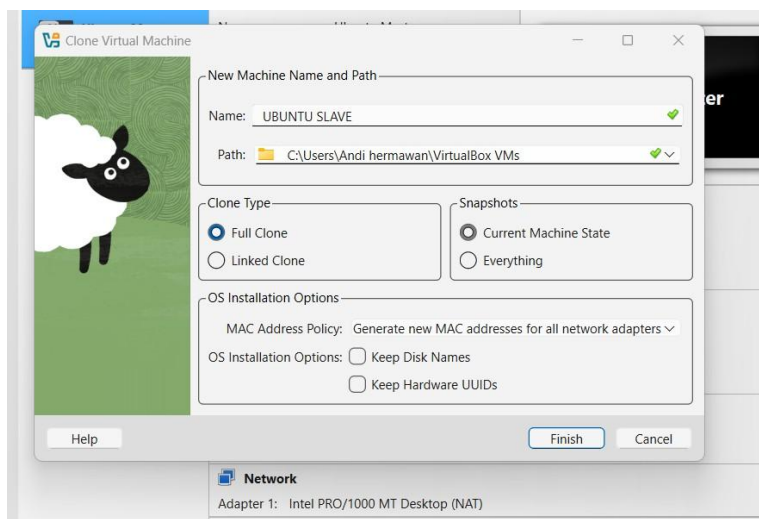
Copyright (c) 2000, 2023, Oracle and/or its affiliates.

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> _
```

Membuat Clone dengan nama UBUNTU SLAVE:



Mulai ulang change slave:

```
mysql> STOP SLAVE;
Query OK, 0 rows affected (0.00 sec)

mysql> RESET SLAVE ALL;
Query OK, 0 rows affected (0.02 sec)

mysql> CHANGE MASTER TO
  -> MASTER_HOST='192.168.56.101',
  -> MASTER_USER='repl',
  -> MASTER_PASSWORD='password123',
  -> MASTER_LOG_FILE='mysql-bin.000006',
  -> MASTER_LOG_POS=154;
Query OK, 0 rows affected, 2 warnings (0.06 sec)

mysql> START SLAVE;_
```

```
inet6 fe80::a00:27ff:fe0a:4942/64 scope link
valid_lft forever preferred_lft forever
andi@ubuntu-slave:~$ mysql -h 192.168.56.101 -u repl -p -e "SELECT @@server_id, @@hostname;"
Enter password:
ERROR 1045 (28000): Access denied for user 'repl'@'192.168.56.102' (using password: YES)
andi@ubuntu-slave:~$ mysql -h 192.168.56.101 -u repl -p -e "SELECT @@server_id, @@hostname;"
Enter password:
+-----+
| @@server_id | @@hostname |
+-----+
| 1 | ubuntu-master |
+-----+
andi@ubuntu-slave:~$
```

PENJELASAN PRAKTIKUM:

Praktikum replikasi MySQL diawali dengan menyiapkan dua mesin virtual di VirtualBox, yaitu Ubuntu yang berperan sebagai Master dan Ubuntu sebagai Slave. Pada kedua VM tersebut, MySQL telah berhasil dipasang dan dikonfigurasi sesuai kebutuhan. Pengaturan yang dilakukan meliputi server-id, aktivasi binary log pada server Master, serta pembuatan akun repl yang digunakan khusus untuk proses replikasi. Tahap awal instalasi dan konfigurasi dapat diselesaikan dengan baik, namun kendala mulai muncul saat melakukan pengaturan pada sisi Slave. Salah satu masalah yang ditemui adalah error SUPER privilege ketika menjalankan perintah CHANGE MASTER TO, sehingga perintah tersebut harus dieksekusi menggunakan akun root. Setelah itu, muncul kembali error 1045 Access Denied yang menunjukkan bahwa Slave tidak dapat melakukan autentikasi ke server Master.

Permasalahan ini cukup membingungkan karena saat dilakukan uji login secara manual dari Slave ke Master menggunakan akun repl, koneksi dapat berhasil. Akan tetapi, ketika digunakan dalam proses replikasi otomatis, koneksi tetap gagal. Hal tersebut mengindikasikan adanya perbedaan mekanisme autentikasi antara login manual dengan autentikasi yang dijalankan oleh sistem replikasi MySQL. Selain itu, hambatan lain juga berasal dari konfigurasi jaringan pada VirtualBox. Pada awalnya, mode NAT membatasi komunikasi antar mesin virtual. Setelah pengaturan jaringan diubah dan koneksi diuji menggunakan perintah ping, barulah kedua VM dapat saling terhubung dengan baik.